EXERCISE

1. Basic Greeting Script:
    - Create a script named welcome.sh that prints a simple welcome message and then on a new line, displays the output of the pwd command.

```
hp@TF-COMP-786:~/my_project$ cat welcome.sh
#!/bin/bash
echo "Welcome to the Linux scripting exercise!"
pwd
hp@TF-COMP-786:~/my_project$ chmod +x welcome.sh
hp@TF-COMP-786:~/my_project$ ./welcome.sh
Welcome to the Linux scripting exercise!
/home/hp/my_project
```

2. Using Variables:
    - Create a script named user_info.sh.
    - Inside the script, create a variable named USERNAME and set its value to your own name.
    - The script should print a message like: "The script owner is: [Your Name]".

```
hp@TF-COMP-786:~/my_project$ cat > user_info.sh
#!/bin/bash
USERNAME="Tooba Fayyaz"
echo "The script owner is: $USERNAME"
hp@TF-COMP-786:~/my_project$ chmod +x user_info.sh
./user_info.sh
The script owner is: Tooba Fayyaz
```

3. Using Positional Parameters:
    - Write a script named file_creator.sh that accepts one argument: a filename.
    - The script should create an empty file with the name provided as the argument.
    - The script should then print a confirmation message: "File '[filename]' has been created."
    - (Hint: Use the touch command and the $1 variable).

```
hp@TF-COMP-786:~/my_project$ cat > file_creator.sh
#!/bin/bash
# Check if a filename was provided
if [ -z "$1" ]; then
  echo "Usage: $0 <filename>"
  exit 1
fi

touch "$1"
echo "File '$1' has been created."
hp@TF-COMP-786:~/my_project$ chmod +x file_creator.sh
./file_creator.sh myfile.txt
File 'myfile.txt' has been created.
```

4. Using Built-in Variables:

- Write a script named arg_check.sh.
- The script should print the following information:
  - "The script's name is: [script name]"
  - "You provided [number] arguments."
- (Hint: Use the $0 and $# variables).

```
hp@TF-COMP-786:~/my_project$ cat > arg_check.sh
#!/bin/bash
echo "The script's name is: $0"
echo "You provided $# arguments."
hp@TF-COMP-786:~/my_project$ chmod +x arg_check.sh
./arg_check.sh one two three
The script's name is: ./arg_check.sh
You provided 3 arguments.
```

5. Putting It All Together:
   - Create a script named dir_report.sh that takes a directory path as its only argument.
   - The script should perform the following actions: 1. Print a message: "--- Generating report for directory: [directory path] ---" 2. Create a variable that stores the number of items in that directory. (Hint: use command substitution with a pipeline like ls [directory path] | wc -l). 3. Print another message: "The directory contains [number] items."

```
hp@TF-COMP-786:~/my_project$ cat > dir_report.sh
#!/bin/bash

# Check if directory path provided
if [ -z "$1" ]; then
  echo "Usage: $0 <directory_path>"
  exit 1
fi

DIRECTORY="$1"

echo "--- Generating report for directory: $DIRECTORY ---"

ITEM_COUNT=$(ls "$DIRECTORY" | wc -l)

echo "The directory contains $ITEM_COUNT items."
hp@TF-COMP-786:~/my_project$ chmod +x dir_report.sh
./dir_report.sh /home/hp
--- Generating report for directory: /home/hp ---
The directory contains 14 items.
```